Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \sum_j q_a^j(|X_j|^2 - |Y_j|^2)\eta_j^+\overline{\eta}_j^+ &= r_a^2\eta_a^+\overline{\eta}_a^+ && (a) \\ \sum_j q_a^j(X_j\overline{Y}_j)\eta_j^+\eta_j^+ &= 0 && (b) \\ \sum_j q_a^j(\overline{X}_jY_j)\overline{\eta}_j^+\overline{\eta}_j^+ &= 0. && (c) \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \tilde{\nu}_v \cdot \psi &= \left(\frac{g_u \times \psi}{|g_u \times \psi|}\right)_v \cdot \psi \\ &= \left(\frac{g_{uv} \times \psi + g_u \times \psi_v}{|g_u \times \psi|}\right) \cdot \psi + \big((g_u \times \psi) \cdot \psi\big)\left(\frac{1}{|g_u \times \psi|}\right)_v \\ &= -\frac{\det(g_u, \psi, \psi_v)}{|g_u \times \psi|} \neq 0. \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{cases} p_t + u \cdot \nabla p + \gamma p \nabla \cdot u = 0, \\ beginalign*6pt]\tau^2 u_t + \tau^2 u \cdot \nabla u + \frac{1}{\varrho}\nabla p + u = 0, \\ beginalign*6pt]S_t + u \cdot \nabla S = 0, \\ beginalign*6pt](p, u, S)(x, 0) = (p_0(x,\tau), \frac{u_0(x,\tau)}{\tau}, S_0(x,\tau)), \end{cases}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\Delta_{2k}\left(f(x), g(x)\right) = \det \begin{pmatrix} a_0 & a_1 & a_2 & \dots & a_{2k-1} \\ b_0 & b_1 & b_2 & \dots & b_{2k-1} \\ 0 & a_0 & a_1 & \dots & a_{2k-2} \\ 0 & b_0 & b_1 & \dots & b_{2k-2} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \dots & b_k \end{pmatrix}_{2k \times 2k} .$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$L_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$